

RISKSAARTHI – Test Case Documentation

1. Overview

Project Name: RISKSAARTHI

Module Tested: Loan Risk Prediction System

Testing Type: Unit, Integration, Functional, Validation, Performance

Tools Used: pytest, FastAPI TestClient

2. Test Cases

TC-01: Model Info API

Field	Value
Test Case ID	TC-01
Endpoint	/api/model-info
Input	None
Expected Output	Model name, version, last_updated
Actual Result	PASS
Status	PASS

TC-02: Valid Loan Prediction

Field	Value
Test Case ID	TC-02
Endpoint	/api/predict
Input	Valid payload
Expected Output	Prediction + probability + decision
Actual Result	PASS
Status	PASS

TC-03: Invalid Payload

Field	Value
Test Case ID	TC-03
Input	Missing/incorrect fields
Expected Output	400 / 422 error
Actual Result	PASS
Status	PASS

TC-04: Boundary Values

Field	Value
Test Case ID	TC-04
Input	Minimum valid values
Expected Output	Successful prediction
Actual Result	PASS
Status	PASS

TC-05: Invalid Identity Format

Field	Value
Test Case ID	TC-05
Input	Invalid PAN/Aadhaar
Expected Output	400 error
Actual Result	PASS
Status	PASS

TC-06: Identity Security Check

Field	Value
Test Case ID	TC-06
Input	Valid request
Expected Output	Identity number NOT returned
Actual Result	PASS
Status	PASS

TC-07: Database Failure Handling

Field	Value
Test Case ID	TC-07
Input	Simulated DB failure
Expected Output	API should handle gracefully
Actual Result	PASS
Status	PASS

TC-08: Prediction Consistency

Field	Value
Test Case ID	TC-08
Input	Same payload twice
Expected Output	Same prediction result
Actual Result	PASS
Status	PASS

TC-09: Preprocessing Validation

Field	Value
Test Case ID	TC-09
Input	Missing optional fields
Expected Output	Preprocessing still works
Actual Result	PASS
Status	PASS

TC-10: Performance Test

Field	Value
Test Case ID	TC-10
Input	Valid payload
Expected Output	Response within acceptable time
Actual Result	PASS
Status	PASS

TC-11: Explainability Output

Field	Value
Test Case ID	TC-11
Input	Valid data
Expected Output	Explanation string/list
Actual Result	PASS
Status	PASS

TC-12: Database Insert Logic

Field	Value
Test Case ID	TC-12
Input	Valid application data
Expected Output	Record inserted
Actual Result	PASS
Status	PASS

TC-13: Logging Validation

Field	Value
Test Case ID	TC-13
Input	Prediction request
Expected Output	Logs generated
Actual Result	PASS
Status	PASS

3. Test Coverage Summary

Category	Tests Count	Test Case Count
API Tests	9	5
Model Tests	4	2
Utils Tests	4	2
DB Tests	2	1
Performance	1	1
Security	1	1
Logging	1	1

Category	Tests Count	Test Case Count
Data Generation	2	1
Explanation	2	1

4. Summary

Category	Status
Unit Testing	Completed
Integration Testing	Completed
Functional Testing	Completed
Validation Testing	Completed
Performance Testing	Completed
Security Testing	Completed
Logging Testing	Completed

5. Final Result

1. All critical components tested
2. System stable and reliable
3. Ready for deployment / production use